

CHEMISTRY EXAM-23 — DETAILED MARKING REPORT

Chapter 10 (Acids, Bases & Salts, 70%) + Revision: Chapters 4, 6, 8, 9 (30%)

Seemab · Class 9 · Attempt Date: 2026-09-21 · Paper: 50 marks, 2h15m, no optional questions

Final Score: 36 / 50 · 72% · Grade B

CORRECTED after a blind deep check (2026-09-21). The first-pass reading of two items was wrong and both were resolved directly against the scan during the independent deep check: (1) in Q.2(ii) she wrote "H30 { bronsted acid }" (a clear subscript 3), not "H2O" as first read — her error is naming the product H3O+ as the acid instead of the proton donor HCl, and water-as-base was only missed once on this paper (MCQ 1), not twice. (2) In Q.9(ii) and (iii), the products line is genuinely drawn **lower** than the reactants line in both diagrams — the first pass misread them as level. Both corrections raise the score from 35.5 to 36/50. See the judgement-call table for the full pricing rule applied.

Version attempted: VERSION B. Her Q.8 is "Sodium:- $4\text{Na}_2 + \text{O} \rightarrow 2\text{Na}_2\text{O}$... magnesium:- $2\text{Mg} + \text{O} \rightarrow 2\text{MgO}$ " followed by Na2O/CaO/MgO with water and a neutralization part — this is Version B's "Oxides, hydroxides and neutralization" question, not Version A's acid rain question. Marked against answers . pdf (Version B key). The Version A generous rule on Q8(iv) acid rain does not apply.

9 / 10

SECTION A (MCQS)

18.5 / 24

SECTION B (SHORT)

8.5 / 16

SECTION C (LONG)

NEW CHAPTER (Ch 10) — A Q1–7, B Q2–6, C Q8
24.5 / 35 (70.0%)

REVISION (Ch 4, 6, 8, 9) — A Q8–10, B Q7, C Q9
11.5 / 15 (76.7%)

SECTION A — Multiple Choice Questions (9 / 10)

#	Question (summary)	Correct	Seemab's	Result
1	In $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$, water behaves as a	C — Bronsted acid	A — Bronsted base	0
2	Base but not an alkali	B — $\text{Cu}(\text{OH})_2$	B	1
3	Definition of a strong acid	A — dissociates completely	A	1
4	Gas from marble + dil. H_2SO_4	D — carbon dioxide	D	1
5	Gas from $\text{NH}_4\text{Cl} + \text{NaOH}$	A — ammonia	A	1
6	Every acid solution contains excess of	B — H^+ ions	B	1
7	Ordinary rain is slightly acidic because of	C — carbon dioxide	C	1
8	[Revision] Catalyst for margarine manufacture	D — nickel	D	1 RE-TEST FIXED
9	[Revision] Config 2,8,2 belongs to	A — Group 2, Period 3	A	1 RE-TEST FIXED
10	[Revision] At dynamic equilibrium, forward rate is	B — equal to reverse rate	B	1 RE-TEST FIXED

Only Q1 wrong — re-checked directly on the scan (zoomed crop), her bubble clearly reads "A", not a scanning artifact. All three re-tested MCQs (8, 9, 10) are fixed. This is her only water-as-base slip on the paper: she calls water the Bronsted base here when it is actually the acid in this specific reaction. Q.2(ii) below is a related but distinct error (misidentifying which species is the acid in a different reaction), not a repeat of this same mistake.

SECTION B — Short Questions (18.5 / 24)

Q.2(i) — Define Bronsted-Lowry acid and base 2 / 2

Seemab wrote: "An acid is proton donor that means an acid donates protons" .

"A base is Proton acceptors. that means a base accepts protons"

Both definitions correct and complete.

Q.2(ii) — $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$: identify acid/base with reason 1 / 2

Seemab wrote (verbatim, in order):

"H30 { bronsted acid }"

"Reason:- HCl is an acid because it donates protons. whereas H2O is base, and that's why H2O is acids because HCl accepts, donates protons and H2O accepts a protons."

Her label is **H3O** (the product, hydronium), not H2O or HCl — a clear subscript 3, confirmed at high zoom against the scan. Her error is naming the acidic-looking product as the Bronsted acid instead of the proton donor HCl. Credit: 1 mark for correctly identifying H2O as the base ("H2O is base"); 0 for the acid, since neither "H3O" (her label) nor the muddled restatement correctly names HCl as the acid. The reasoning also briefly contradicts itself ("H2O is acids... HCl accepts") straight after correctly stating "H2O is base" — a labelling-discipline issue on top of the content error.

Q.3 — Ionization equations + weak acid ID 1.5 / 4

(i) HNO₃ — Seemab wrote: "H + NO₃ → HNO₃" 0/1

This is the reverse reaction (ions combining to form the acid), not ionization (the acid splitting into ions). The key wants HNO₃(l) → H⁺(aq) + NO₃⁻(aq). No charges or state symbols either.

(ii) H₂SO₄ — Seemab wrote: "H₂ + SO₄ → H₂SO₄" 0/1

Same reversed-direction error, and "H₂" is molecular hydrogen, not the H⁺ ion required.

(iii) CH₃COOH — Seemab wrote: "CH₃COOH ⇌ H₂ + CHCOO" 0.5/1

Direction is correct (acid → ions) and she used the reversible arrow (⇌) correctly for a weak acid, but the products are wrong: "H₂" instead of H⁺, "CHCOO" instead of CH₃COO⁻, and no charges anywhere. Half credit for the correct arrow and direction.

(iv) Weak acid + reason — Seemab wrote: "it is a weak acid because it do not ionizes completely" 1/1

Correct identification and correct reason.

Q.4 — Balanced equations with state symbols 3 / 4

(i) Zn + HCl — Seemab wrote: "Zn(s) + 2HCl(aq) → ZnCl₂(aq) + H₂(g)", gas released "= hydrogen gas" 2/2

Fully correct, fully balanced, and every state symbol present — this is the exact instruction (HIDAYAAT #6) she has never followed before on a Chemistry paper.

(ii) Na₂CO₃ + HCl — Seemab wrote: "Na₂CO₃(s) + 2HCl(aq) → NaCl(aq) + H₂O(l) + CO₂(g)", gas released "CO₂" 1/2

State symbols present and gas correctly named (½ + ½), but the equation is not balanced: 2 Na atoms on the left need 2NaCl on the right; she wrote plain "NaCl". Lost the 1-mark balancing point only.

Q.5 — Alkalis vs bases 4 / 4

(i) "All alkalis are bases bcuz alkalis are bases that dissolve in water and produce OH ions whereas Not all bases are alkalis bcuz some bases do not dissolve in water."

2/2

Full, complete answering sentence — a good example of HIDAYAAT #7 being followed correctly.

(ii) Alkali: "NaOH" · Not an alkali: "CuO" 2/2

CuO is not in the key's example list (Cu(OH)₂, Al(OH)₃, Fe(OH)₃) but it is scientifically valid: the chapter defines bases as oxides OR hydroxides of metals, and CuO is an insoluble metal oxide, so it is a base that is not an alkali. Credited in full — the key's example list was illustrative, not exhaustive.

Q.6 — Properties of bases + KOH/NH₄Cl 3.5 / 4

(i) "1. Bases change the colour of red litmus paper to blue. 2. Base neutralizes acids to form salt and water." 2/2

(ii) "KOH(aq) + NH₄Cl(aq) → KCl(aq) + H₂O(l) + NH₃(g)" 1.5/2

Correctly balanced with full state symbols (1 mark), but the gas is never named in words separately from the equation — NH₃(g) appearing inside the equation is not the same as writing "gas liberated: ammonia" as the question asks. Half mark lost for the missing name, priced the same as every other missing-name/unit item on this paper.

Q.7 — [Revision] 0.2 mol NaOH calculations 3.5 / 4

(i) "23+16+1 = 40 g/mol" 1/1

(ii) "mass = m.m x mole = 40 x 0.2 = 8" 1.5/2

Correct method and correct number, but the final answer is written as a bare "8" with no unit. HIDAYAAT #5 explicitly requires a unit on every numerical answer; half mark lost, priced the same as every other missing-name/unit item on this paper. **This is where the missing-conclusion habit shows up on this paper** — not across all of Section B (Q5(i) ends with a complete sentence, and Q3 carries short bracketed reasons), but specifically here: no sentence such as "She must weigh out 8 g of NaOH" closes the calculation.

(iii) "formula units = 0.2 x 6.022x10²³ = 1.2044x10²³" 1/1

Correct method and correct value (she used the fuller 6.022×10²³ rather than the key's rounded 6.02×10²³, which is not an error).

SECTION C — Long Questions (8.5 / 16)

Q.8 — Oxides, hydroxides and neutralization 3.5 / 8

(i)(a) Sodium — "4Na2 + 0 → 2Na2O", oxide named as "2Na2O"

(i)(b) Magnesium — "2Mg + 0 → 2MgO", oxide name left blank

Equations + names (both equations, both names) 0/3

Both equations are missing oxygen's diatomic form (O2) — she wrote plain "O" where the burning reaction needs "O2". The correct pair is $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ and $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. Neither equation is accepted as balanced as written, and neither oxide is named in English (sodium oxide, magnesium oxide) — for sodium she only repeats the formula a second time, and for magnesium the name line is blank. No marks awarded for this part.

(ii)(a) Na2O + H2O — "Na2O + H2O → 2Na2OH" (the subscript 2 on "Na2OH" is visibly blotted out on the page, a deletion), restated cleanly as

"hydroxide formation:- 2NaOH" 0.5/1

(ii)(b) CaO + H2O — "CaO + H2O → Ca(OH)2", then "hydroxide formation:- Ca(OH)2" 0.5/1

(ii)(c) MgO + H2O — "MgO + H2O → Mg(OH)2", then "hydroxide formation:- Mg(OH)2" 0.5/1

All three equations are correct once the deletion in (a) is read as intended (2NaOH). But in no part does she write the English name (sodium/calcium/magnesium hydroxide) — only the formula, repeated a second time. **Pricing rule applied here:** the key itself prices a missing name/unit inconsistently across this paper (½ mark in Q4, a full mark in Q6, and a whole equation-mark in Q8), which is our defect, not hers; resolved in her favour by pricing every missing name/unit at a flat ½ mark. That gives ½ credit per part here (equation right, name missing) rather than 0.

(iii) Neutralization — "general equation:- Acid + Base → Salt + H2O", example "HCl + NaOH → NaCl + H2O.",

"always formed:- Salt + water." 2/2

Full marks — general equation, one correct balanced example, and the "always formed" statement are all present and correct.

Q.9 — [Revision] Energy changes in a reaction 5 / 8

(i) Define activation energy —

"The minimum amount of energy that in addition to the average kinetic energy where particles must have an effective collisions is called activation energy."

2/2

Both required ideas present: a minimum energy, on top of average KE, needed for an effective collision. Matches her book's own wording.

(ii) Labelled energy profile diagram 2/4

Y-axis labelled "ENERGY" but the x-axis is never labelled "reaction pathway"/"reaction progress" (½/1 on the axes point). **Products ARE drawn clearly lower than reactants** in her diagram — confirmed on close re-reading of the scan — so the levels point is awarded in full (1/1). The hump is drawn and labelled "activation energy", but the label sits at the peak rather than as an arrow running from the reactant level up to the peak, so only half credit for the Ea point (½/1). **ΔH is never labelled or mentioned anywhere** on the diagram (0/1).

(iii) Catalysed pathway 1/2

She does draw a second, lower hump on the same axes, labelled "activation energy without catalyst" for the taller curve and "activation energy with products" for the lower one (she means "with catalyst") — the catalysed curve is genuinely present and correctly drawn lower, so full credit for the drawing (1/1). But nowhere does she write the required sentence that **Ea is lowered while ΔH stays exactly the same** (0/1 — no such statement appears anywhere).

RE-TEST OUTCOMES (her documented open weaknesses, targeted deliberately)

Re-test	Result	Evidence
MCQ 8 — catalyst list, nickel for margarine (missed exam-21, exam-22)	FIXED	Answered "D" (nickel), correct.
MCQ 9 — config 2,8,2 → Group 2 Period 3 (exam-22 gave Group 8, the "+10" slip)	FIXED	Answered "A" (Group 2, Period 3), correct.
MCQ 10 — dynamic equilibrium, forward = reverse rate (missed on OPF school paper)	FIXED	Answered "B" (equal to reverse rate), correct.
Section B Q4 & Q6 — state symbols carry their own half marks (never followed before)	FIXED	Every (s)/(aq)/(l)/(g) is correct in both Q4(i)/(ii) and Q6(ii).
Section B Q7 — full concluding sentence in the calculation (missing-conclusion habit)	NOT FIXED	Q7(ii) ends on a bare "8" with no unit and no sentence. (Elsewhere on this paper the habit is fine: Q5(i) closes with a complete sentence and Q3 carries short bracketed reasons.)
Section C Q9 — Ea labelled + catalysed curve drawn + state ΔH unchanged (exam-22 had neither label nor catalysed curve)	HALF FIXED	"Activation energy" is labelled and a genuine second, lower catalysed curve is drawn (both of exam-22's misses are fixed) — but ΔH is never labelled or discussed, and no sentence states that the catalyst leaves ΔH unchanged.

JUDGEMENT CALLS — range of defensible totals

#	Item	Awarded	Note
1	Q.2(ii) — H3O mislabelled as the acid instead of HCl; H2O correctly called the base	1/2	Content error, not a reading error
2	Q.3(i)/(ii) — ionization equations written in reverse; Q.3(iii) right arrow/direction, wrong ion formulas	1.5/4	Direction/arrow credited where earned
3	Q.8(i) — missing O2 (diatomic oxygen) on both burning equations, no names	0/3	Equations not accepted as balanced
4	Q.8(ii) — equations correct (incl. the blotted-out correction in (a)), names missing throughout	1.5/3	½ per part under the flat pricing rule below

5	Q.9(ii) — products correctly drawn lower than reactants; Ea labelled but not as an arrow; no ΔH	2/4	Corrected from first-pass misread of "same height"
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Pricing rule (question-defect-favours-student): the answer key itself prices a missing name or unit inconsistently across the paper — ½ mark in Q4, a full mark in Q6, and a whole equation-mark in Q8(ii). That inconsistency is the paper's defect, not Seemab's, so it is resolved in her favour: every missing name or unit on this paper is priced at a flat ½ mark. **Range for the record: 33.5/50 under the key exactly as written (no consistency correction), 36/50 with the flat-pricing rule applied. 36/50 (72%) is the final mark.**

STRENGTHS AND WEAKNESSES

Strengths: State symbols are now consistently correct across every equation on the paper (Q4, Q6) — a genuine, first-time fix of a long-standing instruction. All three targeted re-test MCQs (8, 9, 10) landed correctly. The reversible-vs-single arrow distinction for weak/strong acids (Q3iii/iv) is understood. Alkali-vs-base reasoning (Q5) and base properties (Q6i) are both clean, full-mark answers, and Q5(i) is a model of the concluding-sentence instruction done right. The energy-profile diagram correctly shows products below reactants and now has a genuine catalysed curve and an Ea label, both missing on exam-22.

Weaknesses: (1) **Dissociation/ionization equations written backwards** — Q3(i) and (ii) both show the combination reaction (ions → acid) instead of ionization (acid → ions). (2) **Naming the acidic product rather than the proton donor as the Bronsted acid** — in Q2(ii) she labels H3O+ (the product) as the acid instead of HCl (the actual proton donor). (3) **Formula given where a name is asked** — in both Q.8 and Q.6(ii) she supplies a correct chemical formula but never writes the English name/word the question asked for. (4) **Diatomic oxygen** — both Q8(i) burning equations write "O" instead of "O2". (5) **ΔH omitted from the energy diagram** — present nowhere in either Q9(ii) or (iii), despite being asked for by name. (6) **Concluding sentence still missing in the calculation** — Q7(ii) ends on a bare "8"; this is now localized to calculations rather than a whole-Section-B habit.

COMPARISON — RECENT CHEMISTRY ATTEMPTS

Exam	Date	Score	%	MCQ	Note
exam-21-ch1-9	9 Sep 2026	41/46	89.1%	7/12	Written work perfect 16/16
exam-22-ch1-9	14 Sep 2026	42/50	84.0%	9/10	6 of 9 re-set errors fixed
exam-23-ch10-rev4-6-8-9	21 Sep 2026	36/50	72%	9/10	First-ever paper on brand-new Ch 10 content (70.0%) plus a deliberate weak-spot revision block (76.7%); this is her first sub-80% Chemistry score, driven almost entirely by first-attempt content (dissociation direction, naming, O2) rather than a re-tested weakness — all three targeted re-tests among the MCQs landed, and state symbols are fixed for good.

This is not a decline in the usual sense: it is the first exam ever on newly-taught material (Ch 10.5, examined for the first time), stacked with a 30% block of her hardest historic weak spots on purpose. The revision half (76.7%) actually out-performed the new-chapter half (70.0%), which is the expected pattern for a first attempt at brand-new content next to a deliberately-targeted re-test block.